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$$\therefore E(X_i) = P(X_i=1) = \frac{1}{i}$$

$$\therefore E(X) = \sum_{i=1}^n \frac{1}{i}$$

X_i 的取值为0或1, 因此 $P(X_i) = P(X_i^2)$

$$\Rightarrow E(X_i^2) = E(X_i) = \frac{1}{i}$$

$$\therefore \text{Var}(X) = E(X^2) - E(X)^2$$

$$= \sum_{i=1}^n \frac{1}{i} - \sum_{i=1}^n \frac{1}{i^2}$$

$$= H(n) - H^2(n)$$

$$\approx \ln n - \frac{\pi^2}{6}$$

